Statement of Research / Interest

Application: Master of Cybersecurity and Threat Intelligence (MCTI)
University of Guelph

Cybersecurity, to me, is not just about preventing breaches. It is about understanding how systems fail, why they fail, and how to design them so that failure does not become compromise. As digital infrastructure becomes more interconnected and cloud-dependent, the attack surface expands in ways that demand deeper technical awareness and structured threat intelligence. My academic training and hands-on professional experience have shaped my interest in backend security, access control systems, and the defense of distributed environments against evolving threats.

I completed my Bachelor of Information Technology at Dedan Kimathi University of Technology, where I built a solid foundation in database systems, computer networks, operating systems, and software engineering. During my studies, I became increasingly interested in how system components interact — how an application communicates with a database, how servers expose services to networks, and how small misconfigurations can create significant vulnerabilities. Courses in networking strengthened my understanding of protocol behavior and segmentation, while database systems deepened my appreciation for transactional integrity and structured access control. These subjects trained me to think analytically about systems rather than viewing them as isolated tools.

Outside the classroom, I invested significant time exploring cybersecurity independently. I worked extensively with Linux environments and security-focused distributions such as Parrot OS, not simply to experiment, but to understand how real-world attacks are structured. I practiced reconnaissance techniques, studied privilege escalation paths, analyzed service misconfigurations, and explored system hardening strategies. This process helped me appreciate cybersecurity from both the offensive and defensive perspectives. It taught me that strong defense requires understanding the mindset and methodology of attackers, not just installing protective tools.

My professional experience has further grounded this understanding. In a healthcare environment, I have been involved in configuring and maintaining access control and time-attendance systems that require reliability and strict authentication enforcement. Working in this setting exposed me to the operational realities of security — where downtime affects real services and misconfigured access can have serious consequences. I participated in ensuring that authentication systems functioned consistently, that role-based access permissions were correctly applied, and that systems remained stable under daily operational demands. This experience reinforced the importance of layered security: combining technical controls, configuration discipline, and policy enforcement to reduce risk.

Through both independent exploration and professional practice, my interest has grown toward structured threat intelligence and secure system architecture. I am particularly interested in how adversaries exploit architectural weaknesses in distributed systems, how logging and telemetry data can be used for early detection, and how secure-by-design principles can be applied at the backend level before vulnerabilities emerge. Rather than treating security as an afterthought, I am motivated by the idea of embedding resilience into system design from the beginning.

The Master of Cybersecurity and Threat Intelligence (MCTI) program at the University of Guelph represents an important step in this progression. What draws me to the program is its integration of technical cybersecurity practices with intelligence analysis and strategic threat assessment. I am especially interested in deepening my understanding of threat modeling, digital forensics, incident response frameworks, and intelligence-driven defense strategies. The program's applied orientation aligns with my desire to translate theory into practice — to move beyond understanding threats conceptually and develop the capability to respond to them systematically.

My immediate goal after completing the MCTI program is to work in industry, particularly in roles focused on security engineering, infrastructure defense, or threat analysis. I want to contribute to environments where secure system design and proactive threat monitoring are critical. In regions where digital transformation is accelerating rapidly, organizations often expand faster than their security maturity. I hope to be part of strengthening that maturity through disciplined architecture, monitoring strategies, and intelligence-informed defense. Over time, as I gain practical exposure to evolving threat landscapes, I remain open to pursuing deeper research engagement, particularly in areas related to distributed system security and adversary behavior analysis.

I approach cybersecurity with persistence, discipline, and a commitment to continuous learning. My academic foundation provided structure, my independent exploration built technical intuition, and my professional experience taught me operational responsibility. Together, these experiences have prepared me for the rigor of graduate study. Through the MCTI program, I aim to refine my analytical thinking, expand my threat intelligence capabilities, and contribute meaningfully to the broader effort of protecting modern digital infrastructure. I am motivated not only by technical challenge, but by the responsibility that comes with securing systems people depend on every day.